

SELF STUDY NOTES

DNS & Internet Architecture

Domain Name System • Network Models • Peer-to-Peer • Cloud • SDN

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Table of Contents

01	What is DNS?	11	Client-Server Pros & Cons
02	Domain Names vs IP Addresses	12	Hybrid Architecture
03	DNS Hierarchy	13	Cloud Architecture
04	DNS Resolution Process	14	Cloud Characteristics
05	DNS Resolution Diagram	15	SDN Architecture
06	Internet Architecture Overview	16	SDN Advantages & Disadvantages
07	Peer-to-Peer (P2P) Architecture	17	Architecture Comparison Table
08	P2P Advantages & Disadvantages	18	Quick Review: DNS Questions
09	Client-Server Architecture	19	Quick Review: Architecture Questions
10	Tier Models in Client-Server	20	Summary & Key Takeaways

"DNS is the phonebook of the Internet"

The Problem

Without DNS, we'd need to memorize complex IP addresses like 93.184.216.34 for every website we visit.

The Solution

DNS lets us type human-friendly names like `www.google.com`. DNS automatically finds the matching IP address.

The Role

DNS bridges the gap between readable domain names and numeric IP addresses required by network hardware.

Domain Names vs IP Addresses

DOMAIN NAME

www.example.com

- ✓ Human-readable
- ✓ Easy to remember
- ✓ Can be meaningful
- ✓ Used in browsers & apps

DNS BRIDGES

IP ADDRESS

93.184.216.34

- ✓ Machine-readable
- ✓ Numerical label
- ✓ Uniquely identifies a device
- ✓ Required by network hardware

 **Key Point:** DNS acts as the translator — you type a name, DNS finds the number. Without DNS, browsing the web as we know it today would be impossible.

DNS Hierarchy

Root Servers

Top of the DNS hierarchy. 13 sets of root servers worldwide.

Top-Level Domains (TLDs)

.com .org .net .uk .de .in

Second-Level Domains

example in example.com

Subdomains / Hostname

www accounts mail blog

Example: www . example . com → Subdomain + Second-Level + TLD

DNS Resolution Process

1

User Input

You type `www.example.com` into your browser.

2

Local Cache Check

Computer checks its local DNS cache for a stored IP.

3

Recursive DNS Server

If not cached, query goes to a Recursive DNS Server (usually from your ISP or Google DNS).

4

Root Server

Recursive server contacts a Root Server, which points to the TLD name server.

5

TLD Name Server

The `.com` TLD server directs the query to the authoritative server for `example.com`.

6

Authoritative Name Server

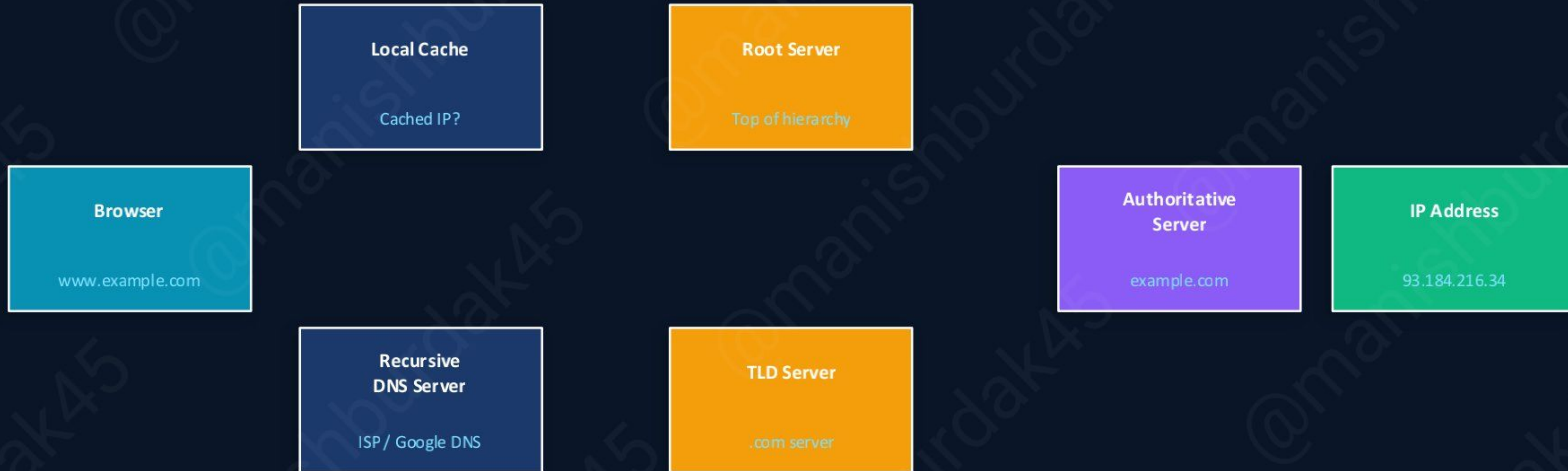
This server returns the actual IP address for `www.example.com`.

7

IP Returned & Connected

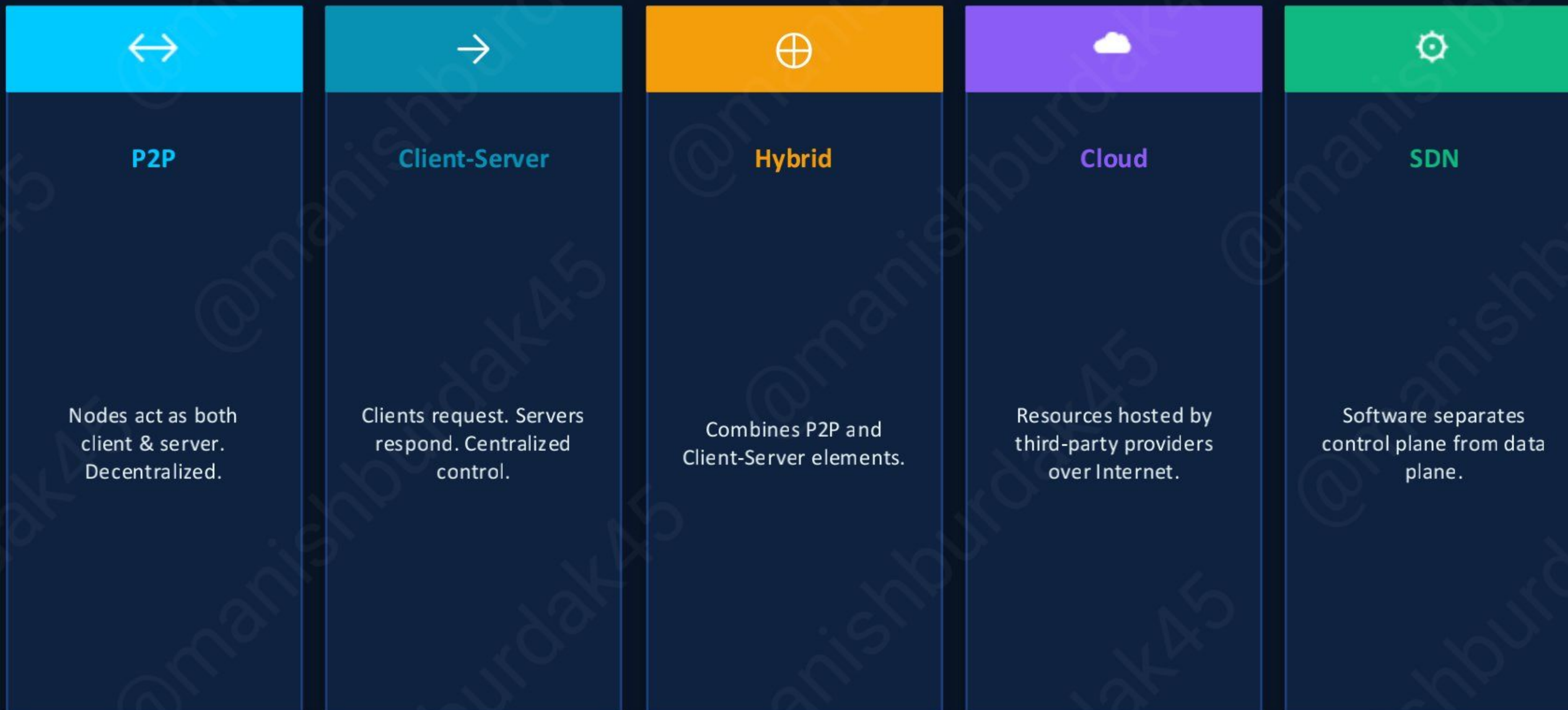
IP is sent back to your computer, which connects directly to the website.

DNS Resolution — Visual Flow



💡 **This entire process happens in milliseconds!** Once the IP is found, your browser connects directly to the web server. Results are cached locally to speed up future visits.

Internet Architecture — Overview



Each architecture has unique trade-offs in scalability, security, cost, and management complexity.

Peer-to-Peer (P2P) Architecture

What is P2P?

In a P2P network, each node (computer/device) acts as BOTH a client AND a server. Nodes communicate directly with each other, sharing resources without needing a central server.

P2P Types

- Fully Decentralized: No central server at all
- Partially Centralized: Central server for some coordination only

Real World Examples:

BitTorrent (seeders upload, leechers download from multiple sources) **Blockchain** (nodes validate transactions without a central authority) **Friends sharing photos**

How P2P File Sharing Works:

1

Install P2P app on each computer

2

Select folder to share with others

3

Everyone performs the same setup

4

Browse & download directly from peers

P2P — Advantages & Disadvantages

✓ Advantages

Scalability

Adding more nodes increases total resources (storage, CPU, bandwidth).

Resilience

If one node goes offline, others continue functioning. No single point of failure.

Cost Distribution

Resource burden (bandwidth, storage) is spread across peers — very cost-efficient.

✗ Disadvantages

Management Complexity

Harder to control and enforce updates & security policies across all nodes.

Potential Reliability Issues

If too many peers leave the network, resources may become unavailable.

Security Challenges

Each node is exposed to potential vulnerabilities; malware can spread easily.



Use Case: P2P is best for file sharing, blockchain, and distributed computing where centralized control is not needed.

Client-Server Architecture

What is Client-Server?

The most widely used architecture on the Internet. Clients (user devices) send requests to centralized servers which respond with data or services. Multiple clients connect to dedicated servers.

Examples

- Websites (browser ↔ web server)
- Email (client ↔ mail server)
- Online games
- Banking apps

Request Flow Example



How it works:

1. Open browser → type weatherexample.com
2. Browser sends request over Internet to the server
3. Server processes request → locates the page → sends back HTML/data
4. Browser renders the webpage for you

Tier Models in Client-Server

Single-Tier

Client

Client, server, and database all on one machine. Simple but NOT scalable. Used for local apps.

Two-Tier

Client

App Server

Client handles presentation; server manages data. Typical in desktop apps. Client talks directly to DB server.

Three-Tier

Client

App Server

Database

Adds an Application Server between client and DB. Client = Presentation, App Server = Business Logic, DB = Data. Most common for web apps.

N-Tier

Client

App Server

Database

...

Multiple application server layers for complex systems. Highly scalable and distributed. Ideal for enterprise web apps.

⚠ Important: In a typical web app, the browser does NOT interact directly with the database — the web server handles that.

Client-Server — Advantages & Disadvantages

✓ Advantages

Centralized Control

Easier to manage, update, and apply security policies from one place.

Security

Central security policies, firewalls, and authentication can be enforced.

Performance

Dedicated servers can be highly optimized for their specific tasks.

✗ Disadvantages

Single Point of Failure

If the central server goes down, ALL clients lose access.

High Cost & Maintenance

Setting up and sustaining servers is expensive, requiring expert management.

Network Congestion

High traffic can slow or disrupt connections when too many clients connect simultaneously.



Use Case: Websites, email services, online banking — anywhere centralized control and security are priorities.

What is Hybrid Architecture?

A Hybrid model blends elements of both Client-Server and P2P architectures. Central servers handle coordination & authentication, while actual data transfer happens directly between peers.

Video Conferencing Example:



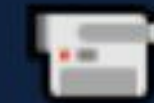
Login

Central server verifies credentials



Session Setup

Server coordinates meeting participants



Video/Audio

Data flows directly peer-to-peer (P2P)



Result

Low lag + central control combined

✓ Advantages

- Efficiency: Relieves server load via P2P data sharing
- Control: Central server still manages auth & directory

✗ Disadvantages

- Complex implementation (centralized + distributed)
- If central coordinating server fails, peer discovery stops

What is Cloud Architecture?

Computing infrastructure hosted and managed by third-party providers (AWS, Azure, Google Cloud). Operates on a virtualized scale, following a client-server model. Users access resources over the Internet without controlling hardware.

Providers

- ☁ Amazon Web Services (AWS)
- ☁ Microsoft Azure
- ☁ Google Cloud Platform
- ☁ Google Drive / Dropbox (SaaS)

SaaS Model Example:

Google Drive / Dropbox — Access applications over the internet without managing the underlying hardware (SaaS = Software as a Service).

✓ Advantages

- Scalability: Easily add/remove computing resources
- Reduced cost: Hardware managed by provider
- Flexibility: Access from anywhere with Internet

✗ Disadvantages

- Vendor lock-in: Migrating providers is complex
- Security/Compliance: Data privacy with third party
- Connectivity: Requires stable Internet access

 *Use Case: Cloud storage, SaaS, PaaS, IaaS — businesses that need scalable, managed infrastructure.*

5 Essential Cloud Characteristics



On-Demand Self-Service

Automatically set up and manage cloud services without human intervention from the provider.



Broad Network Access

Access cloud services from any internet-connected device — laptop, phone, tablet, anywhere.



Resource Pooling

Share and allocate service resources dynamically among multiple users using multi-tenant models.



Rapid Elasticity

Quickly scale services UP or DOWN based on real-time demand. Pay for what you need when you need it.



Measured Service

Only pay for the resources you actually use. Usage is tracked with precision for billing purposes.

These 5 characteristics are the NIST definition of cloud computing — memorize them for exams!

Software-Defined Networking (SDN)

What is SDN?

SDN separates the Control Plane (decides where traffic goes) from the Data Plane (actually forwards traffic). The control plane is centralized in a software controller, making the network programmable.

Use Cases

- Large enterprise networks
- Cloud data centers
- Dynamic bandwidth allocation
- Real-time traffic management

Control Plane

The 'brain'. Makes decisions about where network traffic is directed. Centralized in SDN Controller.

Data Plane

The 'muscle'. Actually forwards the network traffic based on instructions from control plane.

SDN Controller

Central software component. Provides a programmable interface to manage and configure the network.

Traditional Networking:

Control + Data plane both inside each router/switch. Manual config per device.

SDN:

Control plane centralized. Devices just follow instructions. Network managed via software.

SDN — Advantages & Disadvantages

✓ Advantages

Centralized Control

Simplifies network management — one controller manages everything.

Programmability & Automation

Network configurations changed quickly through software instead of manually configuring each device.

Scalability & Efficiency

Dynamically optimize traffic flows for better resource utilization.

✗ Disadvantages

Controller Vulnerability

If the central SDN controller goes down, the entire network may be adversely affected.

Complex Implementation

Requires new skill sets, specialized software, and compatible hardware.

Transition Costs

Moving from traditional networking to SDN requires significant investment and retraining.



Use Case: Large data centers, cloud providers, enterprises needing dynamic, programmable network management.

Architecture Comparison Table

Architecture	Centralized?	Scalability	Management	Typical Use Cases
P2P	Decentralized	High	Complex	File-sharing, Blockchain
Client-Server	Centralized	Moderate	Easier	Websites, Email services
Hybrid	Partially central	Higher than C-S	Complex	Messaging, Video conferencing
Cloud	Provider's infra	High	Outsourced	Cloud storage, SaaS, PaaS
SDN	Control plane	High (policy)	Specialized	Data centers, Enterprises

💡 In practice, modern systems often BLEND multiple architectures to balance performance, scalability, and cost.

Quick Review: DNS Questions

Q1: What type of domain is .com considered?

→ Top-Level Domain (TLD)

Q2: In www.example.com, what is 'example' called?

→ Second-Level Domain

Q3: What is checked first in DNS resolution?

→ Local Cache

Q4: What type of DNS server does an ISP typically provide?

→ Recursive DNS Server

Q5: Which server directs recursive DNS to the TLD server?

→ Root Server

Q6: What numerical label uniquely identifies a device?

→ IP Address

Q7: In accounts.google.com, what is 'accounts'?

→ Subdomain

Quick Review: Architecture Questions

Q1: What architecture allows nodes to act as both client and server?

→ Peer-to-Peer (P2P)

Q4: In which architecture is control plane separated from data plane?

→ Software-Defined Networking (SDN)

Q2: What architecture combines Client-Server and P2P models?

→ Hybrid Architecture

Q5: Which architecture is known for decentralized data sharing?

→ Peer-to-Peer (P2P)

Q3: Which cloud model involves accessing apps without managing infrastructure?

→ SaaS (Software as a Service)

Q6: What model does video conferencing use for coordination + data transfer?

→ Hybrid Architecture

Summary & Key Takeaways

DNS

Translates domain names to IP addresses. Hierarchy: Root → TLD → Second-Level → Subdomain. Resolution checks local cache first, then recursive → root → TLD → authoritative server.

Hybrid

Blends P2P + Client-Server. Central server for auth, P2P for data. Used in video conferencing & messaging apps.

P2P

Nodes act as both client & server. Decentralized. Great for file-sharing & blockchain. Scalable but hard to manage.

Cloud

Hosted by third-party providers. 5 NIST characteristics. SaaS/PaaS/IaaS. Scalable, flexible, but vendor lock-in risk.

Client-Server

Most common web architecture. Centralized servers. Tiers: Single, Two, Three, N-Tier. Good control & security but single point of failure.

SDN

Separates control plane from data plane. Centralized software controller. Programmable & efficient but complex.